TrueNorth on AMD Radeon

Local, cost-free inference for an AI agent framework via ROCm + vLLM

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Application Scenario

KisanSathi — an AI farm advisor built on the TrueNorth agent framework. Farmers describe their crop, land, and problem in natural conversation; the agent extracts 13 structured fields and generates a JSON crop advisory aligned to ICAR guidance.

Real-world scope: 120M+ smallholder farmers in India lack affordable access to expert agronomist advice, but already use WhatsApp daily.

Architecture

```
Farmer (text) -> TrueNorthEngine -> LLMRouter
  |-- extract   -> local Qwen2.5-7B (ROCm)
  |-- converse -> local Qwen2.5-7B (ROCm)
  |-- output    -> local Qwen2.5-7B (ROCm)
  `-- verify    -> Claude (cloud, by design)
-> structured JSON advisory
```

AMD Radeon / ROCm Deployment

GPU	gfx1100, 47.98 GiB VRAM, ROCm 7.2.1
Serving	vLLM 0.16.1 (ROCm build), eager mode
Model	Qwen2.5-7B-Instruct
Memory	14.37 GiB weights + 12.69 GiB KV cache
Integration	TrueNorth's built-in LocalLLMClient — no framework code changes required

Observed Model Quality (reported honestly)

Qwen2.5-7B occasionally wraps JSON output in markdown fences with trailing commentary. TrueNorth's field extractor has a fallback parser that recovers the value without failing the turn — a real, working resilience feature of the framework, not added for this submission.

Routing Config (the only change needed)

```
llm:
  routing:
    extract: qwen-local
    converse: qwen-local
    output: qwen-local
  provider_config:
    local:
      provider: openai_compat
      base_url: http://127.0.0.1:8000/v1
      model: Qwen/Qwen2.5-7B-Instruct
```

Benchmark: Cloud vs. Local ROCm

Same TrueNorth engine, same goal, same conversation — only the LLM routing target changed. Numbers are from the app's own structured latency logging, not estimated.

Routing	Turn 1	Cost
Cloud (Gemini → GPT-4o-mini → Haiku)	85,406 ms	metered
Local ROCm (Qwen2.5-7B, Radeon)	**2,073 ms**	**$0.0000**

~41x
FASTER ON LOCAL ROCm

$0.00
MARGINAL INFERENCE COST

Reproduce

```
git clone https://github.com/onenot8/TrueNorth.git
cd TrueNorth/packages/core && poetry install
vllm serve Qwen/Qwen2.5-7B-Instruct --port 8000
GOAL_YAML=.../kisansathi/goal.yaml \
  poetry run python .../kisansathi/app.py
```

Full environment: ROCm 7.2.1 · Python 3.11.15 · vLLM 0.16.1.dev0 (rocm721) · TrueNorth v0.1.8